



BRAC UNIVERSITY
Department of Computer Science Engineering
MIDTERM EXAMINATION, SPRING 2023
CSE 350: Digital Electronics and Pulse Technique



Total Marks: **60 (20 x 3)**

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_{\gamma}(\text{diode}) = 0.6 \text{ V}$, $V_{\gamma}(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

For the logic circuit in figure 1, assume input voltages of V_A , V_B can be **0V/5V**. Also assume that **100 Ω** and **100 k Ω** value resistors are available to be placed in the following circuit. [20]

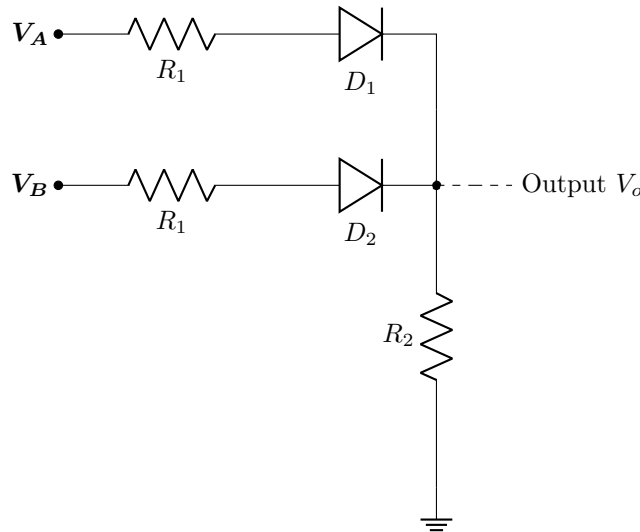


Figure 1

CO2	(a)	From the resistor values available, determine the appropriate resistor values to be used for R_1 and R_2 so that the following circuit acts as a logic OR gate (<i>None of the resistor values can be set to 0</i>). Justify your design by finding the output voltages of the circuit for all possible input voltage combinations.	[10]
CO2	(b)	For your designed circuit , find out the maximum power dissipation .	[10]

Question 2

For the RTL circuit in figure 2, assume $\beta_F = 30$, $V_{BE(saturation)} = 0.8V$, and $V_{CE(saturation)} = 0.2V$. [20]

CO1	(a)	Assume no loads are connected to driver RTL circuit. If so determine the logic gate that the driver circuit represents and justify your answer .	[6]
CO2	(b)	When both inputs are high ($V_A = V_B = 5V$) and no loads are connected to the driver, the total power dissipation in the driver circuit is 220 mW . Estimate the value of R_C in $k\Omega$.	[8]
CO2	(c)	Assume 5 load RTL inverters are connected to the output (V_o) of the driver circuit. Determine the output voltage (V_o) when both inputs are 0 V .	[6]

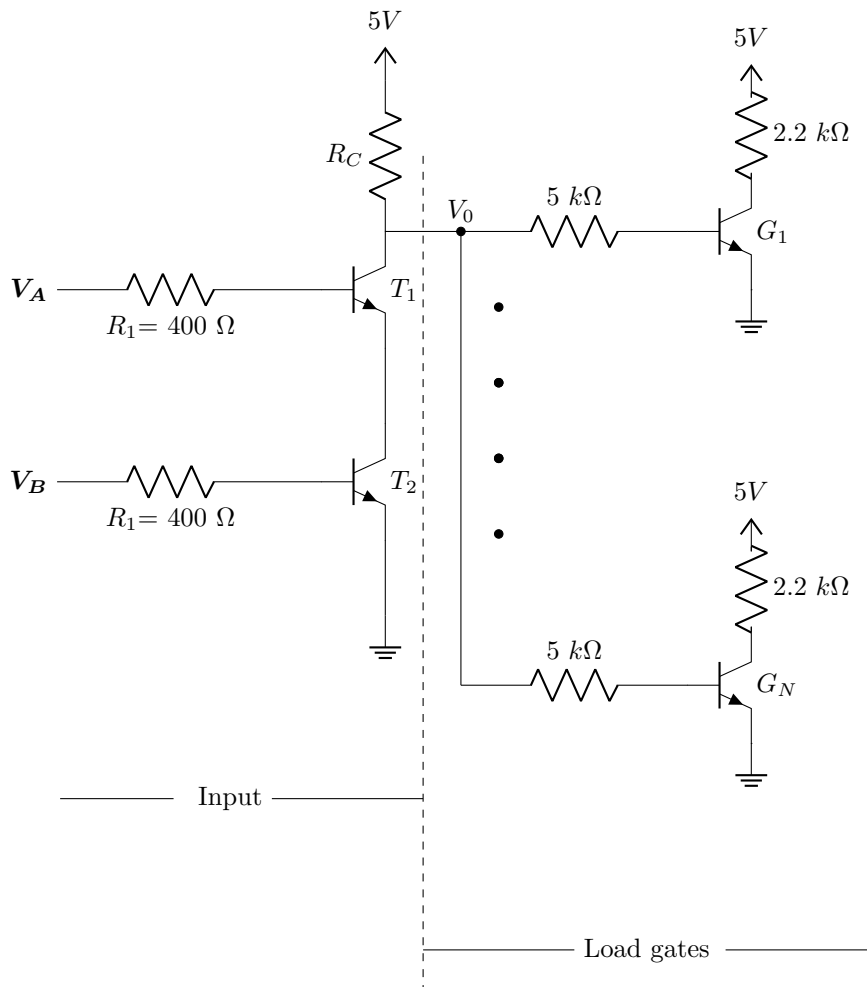


Figure 2

Question 3

For the DTL circuit in figure 3 assume $V_{CE(sat)} = 0.2 \text{ V}$

[20]

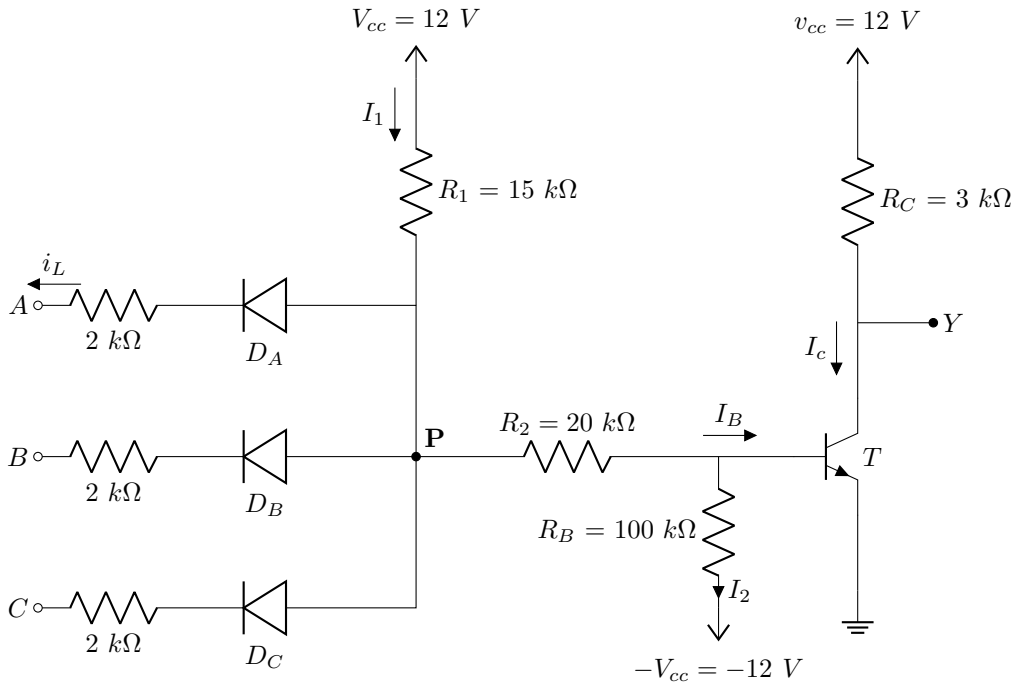


Figure 3

CO1	(a)	What should be the value of logic LOW input of the DTL circuit in the figure 3? Justify the reasoning behind your answer.	[3]
CO2	(b)	Assume inputs A, B, and C have logic LOW values and no load is connected to the output. Find all the currents labeled in the figure. 3.	[8]
CO3	(c)	Verify the operating mode of T transistor when all inputs are high (12V) [<i>maybe activation voltage can help?</i>]	[7]

Question 4

For the given TTL circuit in figure 4, common emitter current gain $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and reverse common emitter current gain of the transistors $\beta_R = 0.1$. Assume both inputs are logical High or 6V. [20]

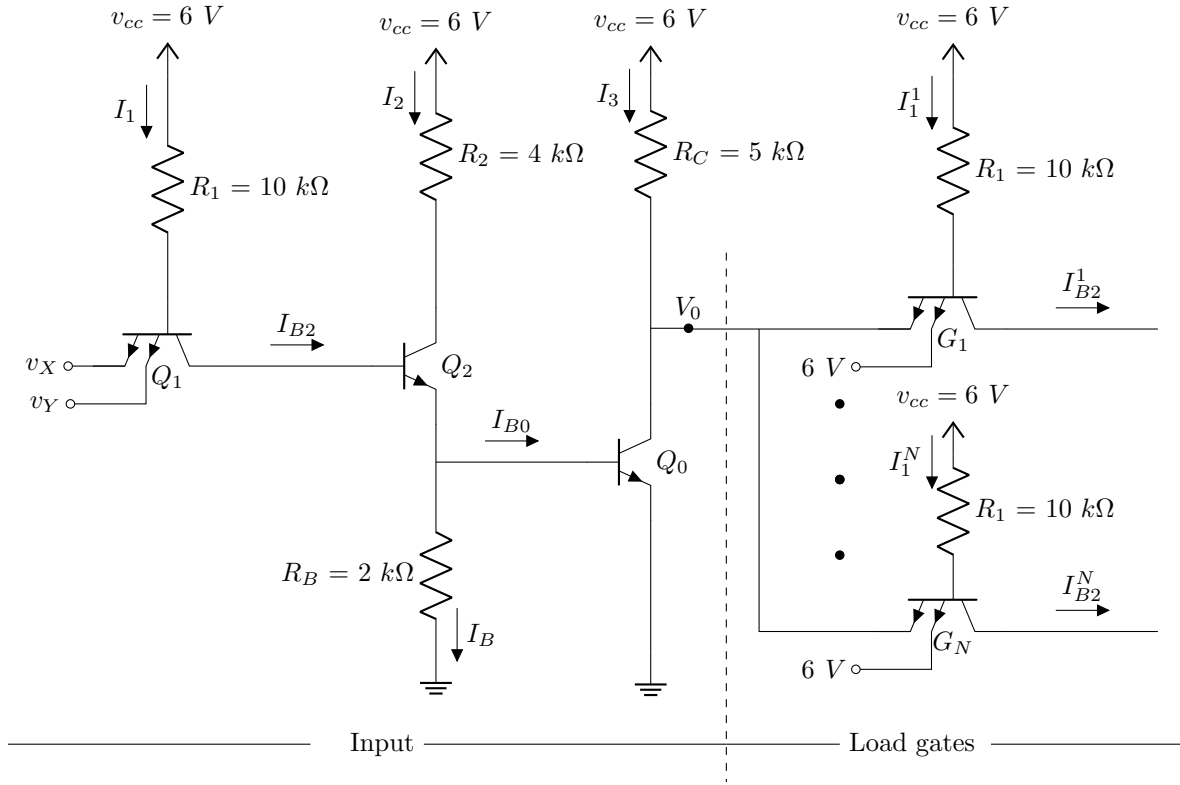


Figure 4

CO1	(a)	Determine the maximum possible fanout for the given input condition only .	[10]
CO2	(b)	Find the total power dissipation in the driver circuit if 10 load gates are connected to the driver.	[6]
CO2	(c)	Select a suitable minimum value of β_F for the bjt Q_0 if we want maximum possible fanout to be 70 .	[4]



Total Marks: **60 (20 x 3)**

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- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

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Question 1

For the DTL circuit in figure 1 assume $V_{CE(sat)} = 0.2 \text{ V}$

[20]

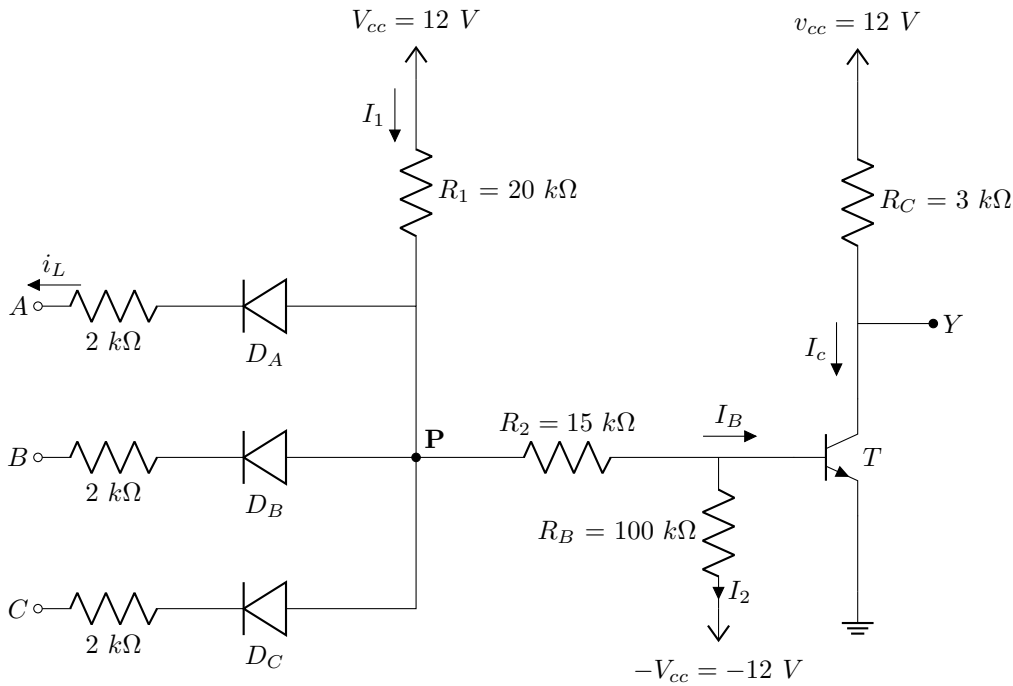


Figure 1

CO1	(a)	What should be the value of logic LOW input of the DTL circuit in the figure 1? Justify the reasoning behind your answer.	[3]
CO2	(b)	Assume inputs A, B, and C have logic LOW values and no load is connected to the output. Find all the currents labeled in the figure. 1.	[8]
CO3	(c)	Verify the operating mode of T transistor when all inputs are high (12V) [<i>maybe activation voltage can help?</i>]	[7]

Question 2

For the given TTL circuit in figure 2, common emitter current gain $\beta_F = 40$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and reverse common emitter current gain of the transistors $\beta_R = 0.1$. Assume both inputs are logical High or 5V. [20]

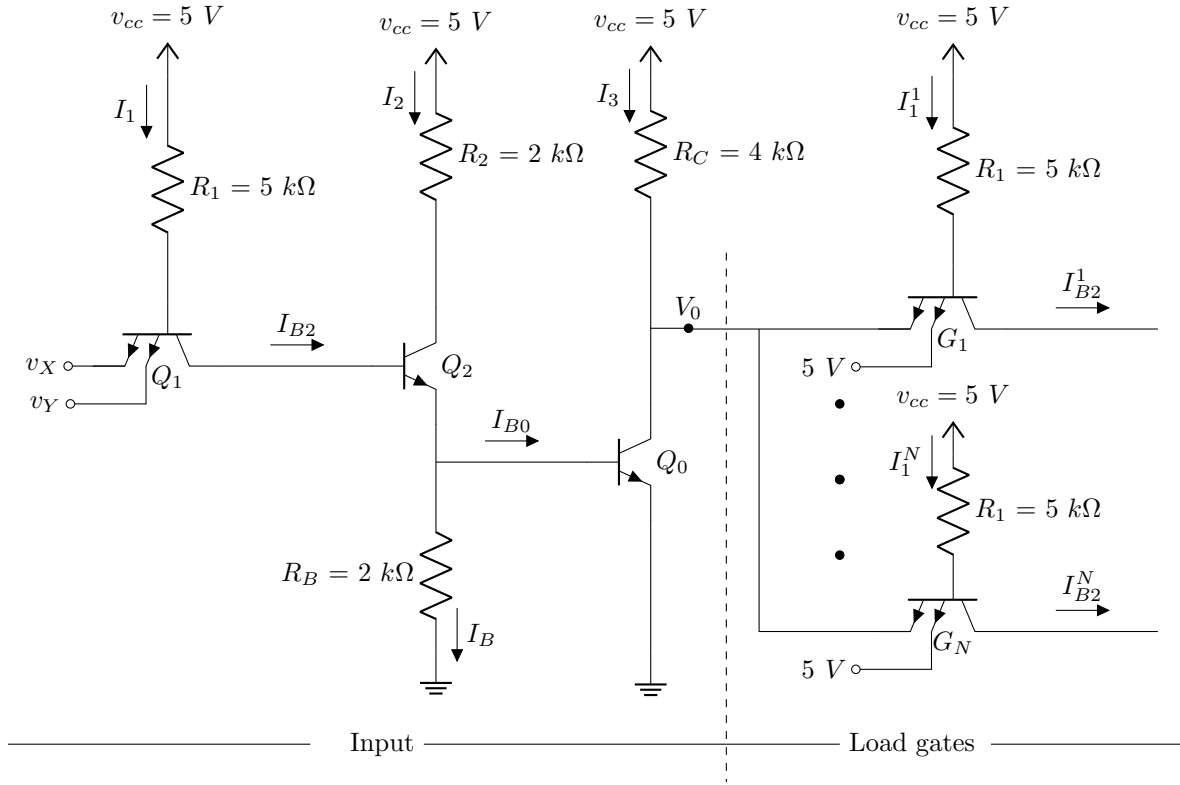


Figure 2

CO1	(a)	Determine the maximum possible fanout for the given input condition only .	[10]
CO2	(b)	Find the total power dissipation in the driver circuit if 20 load gates are connected to the driver.	[6]
CO2	(c)	Select a suitable minimum value of β_F for the bjt Q_0 if we want maximum possible fanout to be 100 .	[4]

Question 3

For the RTL circuit in figure 3, assume $\beta_F = 30$, $V_{BE(saturation)} = 0.8V$, and $V_{CE(saturation)} = 0.2V$. [20]

CO1	(a)	Assume no loads are connected to driver RTL circuit. If so determine the logic gate that the driver circuit represents and justify your answer .	[6]
CO2	(b)	When both inputs are high ($V_A = V_B = 5V$) and no loads are connected to the driver, the total power dissipation in the driver circuit is 175 mW . Estimate the value of R_C in $k\Omega$.	[8]
CO2	(c)	Assume 6 load RTL inverters are connected to the output (V_o) of the driver circuit. Determine the output voltage (V_o) when both inputs are 0 V .	[6]

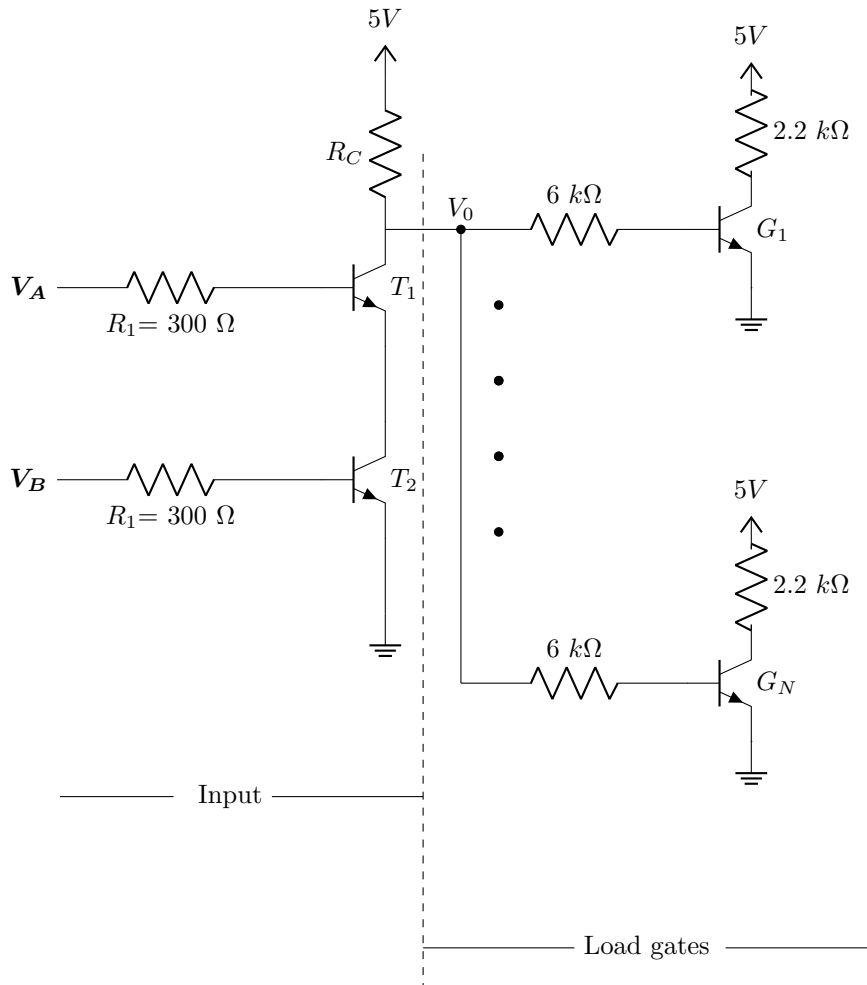


Figure 3

Question 4

For the logic circuit in figure 4, assume input voltages of V_A , V_B can be $0V/5V$. Also assume that $100\ \Omega$ and $100\ k\Omega$ value resistors are available to be placed in the following circuit. [20]

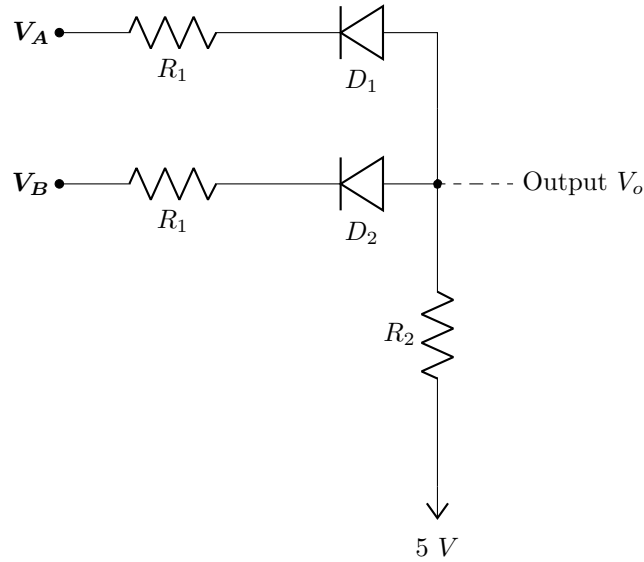


Figure 4

CO2	(a)	From the resistor values available, determine the appropriate resistor values to be used for R_1 and R_2 so that the following circuit acts as a logic AND gate (<i>None of the resistor values can be set to 0</i>). Justify your design by finding the output voltages of the circuit for all possible input voltage combinations.	[10]
CO2	(b)	For your designed circuit , find out the maximum power dissipation .	[10]